

Nils Froleyks

KU Leuven, Belgium

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RESEARCH INTERESTS

Certification, Artificial Intelligence, Automated Reasoning, Hardware Verification, and Planning.

Using formal proofs as a principled way to bridge Artificial Intelligence and Automated Reasoning at scale.

ACADEMIC EXPERIENCE

Postdoctoral Researcher

October 2025 – Present

KU Leuven, Belgium

with Prof. Bart Bogaerts in the Declarative Languages and Artificial Intelligence section

University Assistant

March 2020 – August 2025

Johannes Kepler University Linz, Austria

Teaching duties included courses on Model Checking, Logic, and Debugging.

EDUCATION

Johannes Kepler University Linz, Linz, Austria

March 2020 – August 2025

Ph.D. in Computer Science with Distinction

Thesis: Deep Integration of SAT Solving and Model Checking

Supervisors: Prof. Armin Biere, Prof. Martina Seidl

Karlsruhe Institute of Technology, Karlsruhe, Germany

2018 – 2020

M.Sc. in Computer Science

Best Thesis Award

Thesis: PASAR – Planning as Satisfiability with Abstraction Refinement

Supervisors: Prof. Peter Sanders

Karlsruhe Institute of Technology, Karlsruhe, Germany

2013 – 2017

B.Sc. in Computer Science

Thesis: Using an Algorithm Portfolio to Solve Sokoban

Supervisors: Prof. Peter Sanders

INDUSTRIAL COLLABORATION

Intel Corporation

Nov 2023 – Present

Continued project on integrating hardware model checking certification into industrial workflows.

Woodpecker Technologies, Singapore

2022 – 2023

External consultant. Pre-silicon verification using symbolic formal methods.

CAREER BREAKS

Parental Leave

February – November 2022

AWARDS

CAV Distinguished Paper Award

2026

38th International Conference on Computer Aided Verification

for *Liveness Proofs for Hardware Model Checking*

Fahiem Bacchus PhD Award (Runner-up)

2026

Awarded by the SAT Association for outstanding PhD thesis in the field of satisfiability

Heinz Zemanek Award (Runner-up)

2026

Austrian Computer Society (OCG) award for outstanding PhD theses in computer science

CAV Distinguished Paper Award

2025

37th International Conference on Computer Aided Verification

for *Introducing Certificates to the Hardware Model Checking Competition*

Third Place in Combinatorial Reconfiguration Challenge

2022

CoRe Challenge 2022, Virtual

Best Master's Thesis

2020

Karlsruhe Institute of Technology, Germany

First Prize (Partial Gold Medal)

2019

The Sparkle Planning Challenge, ICAPS, Berkeley, USA

SCIENTIFIC EVENT ORGANIZATION

Organizer, Hardware Model Checking Competition (HWMCC) 2025	FMCAD, Menlo Park, USA
Organizer, Hardware Model Checking Competition (HWMCC) 2024	FMCAD, Prague, Czech Republic
Organizer, SAT Competition 2021	SAT Conference, Barcelona
Organizer, Hardware Model Checking Competition (HWMCC) 2020	FMCAD, Virtual
Organizer, SAT Competition 2020	SAT Conference, Alghero, Italy

SKILLS

Programming Languages: C++, Python
Languages: German (native), English (fluent)

TEACHING & SUPERVISION

Guest Lecturer in Formal Methods for AI Leiden University, Netherlands	Winter Semester 2026
Guest Lecturer in SAT solving KU Leuven, Belgium Advanced course for master's students in computer science.	Summer Semester 2026
Supervised Bachelor's Thesis Leiden University, Netherlands Certifying Constraints from Hardware Designs.	2026
Supervised Bachelor's Thesis Johannes Kepler University Linz, Austria Ternary simulation for IC3 Model Checking.	2024
Lecturer in Debugging Johannes Kepler University Linz, Austria Advanced course for master's students in computer science and artificial intelligence.	Summer Semester 2024
Lecturer in Model Checking Johannes Kepler University Linz, Austria Advanced course for master's students in computer science and artificial intelligence.	Winter Semester 2024
Teaching Assistant in Computer Science and Artificial Intelligence Courses Johannes Kepler University Linz, Austria At both the Master's and Bachelor's levels, hands-on teaching courses include Model Checking, Formal Models, Logic, SAT Solving, Practical Work in AI, and Computational Logics for AI.	2020–2024
Teaching Assistant in Algorithms for Planar Graphs Karlsruhe Institute of Technology, Germany	2018
Teaching Assistant in Theoretical Foundations of Informatics Karlsruhe Institute of Technology, Germany	2015–2019

SOFTWARE

Certifaiger: a certificate checker adopted by the Hardware Model Checking Competition.
Voirag: a certifying hardware model checker.
Cerbtora: a word-level certificate checker for hardware model checking.
CaDiCaL: an award-winning state-of-the-art SAT solver.
CadiBack: a high-performance backbone extractor built on CaDiCaL.

CONTACTS FOR REFERENCES

Prof. Bart Bogaerts, KU Leuven, Belgium. bart.bogaerts@kuleuven.be
Prof. Armin Biere, University of Freiburg, Germany. armin.biere@gmail.com
Prof. Martina Seidl, Johannes Kepler University Linz, Austria. martina.seidl@jku.at

PUBLICATIONS

- [1] **Nils Froleys**, Emily Yu, Bart Bogaerts, Armin Biere, and Keijo Heljanko. *Liveness Proofs for Hardware Model Checking*. International Conference on Computer Aided Verification (CAV), 2026.
Distinguished Paper Award
- [2] Florian Pollitt, Mathias Fleury, Katalin Fazekas, **Nils Froleys**, André Schidler, Dominik Schreiber, and Armin Biere. *CaDiCaL 3.0: Proofs, Algorithms, Implied Literals, and Robust Scheduling*. Theory and Applications of Satisfiability Testing (SAT), 2026.
- [3] **Nils Froleys** and Emily Yu. *Hardware Model Checking Certification with Certifaiger and Cerbtora*. International Joint Conference on Automated Reasoning (IJCAR), 2026.
- [4] Armin Biere, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *Clausal Equivalence Checking*. Journal manuscript under review.
- [5] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Constraints in Hardware Model Checking*. International Symposium on Formal Methods (FM), 2026.
- [6] **Nils Froleys**, Emily Yu, Mathias Preiner, Armin Biere, and Keijo Heljanko. *Introducing Certificates to the Hardware Model Checking Competition*. International Conference on Computer Aided Verification (CAV), 2025.
Distinguished Paper Award
- [7] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL 2.0*. International Conference on Computer Aided Verification (CAV), 2024.
- [8] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal Equivalence Sweeping*. Formal Methods in Computer-Aided Design (FMCAD), 2024.
- [9] Armin Biere, **Nils Froleys**, and Mathias Preiner. *Hardware Model Checking Competition 2024*. Formal Methods in Computer-Aided Design (FMCAD), 2024.
- [10] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal Congruence Closure*. Theory and Applications of Satisfiability Testing (SAT), 2024.
- [11] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Phase Abstraction*. International Joint Conference on Automated Reasoning (IJCAR), 2024.
- [12] **Nils Froleys**, Emily Yu, and Armin Biere. *Ternary Simulation as Abstract Interpretation*. Methoden und Beschreibungssprachen zur Modellierung und Verifikation von Schaltungen und Systemen (MBMV), 2024.
- [13] Armin Biere, **Nils Froleys**, and Wenxi Wang. *CadiBack: Extracting Backbones with CaDiCaL*. Theory and Applications of Satisfiability Testing (SAT), 2023.
- [14] Armin Biere, Mathias Fleury, **Nils Froleys**, and Marijn JH Heule. *The SAT Museum*. Pragmatics of SAT, 2023.
- [15] **Nils Froleys**, Emily Yu, and Armin Biere. *BIG Backbones*. Formal Methods in Computer-Aided Design (FMCAD), 2023.
- [16] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Towards Compositional Hardware Model Checking Certification*. Formal Methods in Computer-Aided Design (FMCAD), 2023.
- [17] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Stratified Certification for K-Induction*. Formal Methods in Computer-Aided Design (FMCAD), 2022.
- [18] **Nils Froleys** and Armin Biere. *Single Clause Assumption without Activation Literals to Speed-up IC3*. Formal Methods in Computer-Aided Design (FMCAD), 2021.
- [19] Tomáš Balyo and **Nils Froleys**. *AI Assisted Design of Sokoban Puzzles Using Automated Planning*. International Conference on ArtsIT, Interactivity and Game Creation, 2021.
- [20] **Nils Froleys**, Marijn Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *SAT Competition 2020*. Artificial Intelligence, 2021.
- [21] **Nils Froleys**, Tomáš Balyo, and Dominik Schreiber. *PASAR—Planning as Satisfiability with Abstraction Refinement*. International Symposium on Combinatorial Search (SoCS), 2019.
- [22] **Nils Froleys** and Tomáš Balyo. *Using an Algorithm Portfolio to Solve Sokoban*. International Symposium on Combinatorial Search (SoCS), 2017.

TECHNICAL REPORTS AND COMPETITION PROCEEDINGS

- [1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, Florian Pollitt, and André Schidler. *CaDiCaL, IsaSAT and Kissat Entering the SAT Competition 2025*. SAT Competition 2025.
- [2] **Nils Froleys**, Emily Yu, and Armin Biere. *Challenging Certificates from Model Checking*. SAT Competition 2025.
- [3] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *Hardware Equivalence Checking Problems Submitted to the SAT Competition 2024*. SAT Competition 2024.
- [4] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL, Gimsatul, IsaSAT and Kissat Entering the SAT Competition 2024*. SAT Competition 2024.
- [5] **Nils Froleys**, Emily Yu, and Armin Biere. *ReconfAIGERation entering Core Challenge 2022*. CoRe Challenge 2022.
- [6] **Nils Froleys**, Emily Yu, and Armin Biere. *Unique Reconfiguration Sequence*. SAT Competition 2022.
- [7] Emily Yu, **Nils Froleys**, Armin Biere, and Mathias Fleury. *Hardware Model Checking Certificates*. SAT Competition 2022.
- [8] Tomáš Balyo, **Nils Froleys**, Marijn JH Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *Proceedings of SAT Competition 2020: Solver and Benchmark Descriptions*. 2020.
- [9] **Nils Froleys**, Tomáš Balyo, and Dominik Schreiber. *PASAR Entering the Sparkle Planning Challenge 2019*. Sparkle Planning Challenge 2019.